

FDR-KG: Supplementary Materials

Code Availability

The complete source code and analysis scripts for all experiments are available at: <https://github.com/Jackxiaozyren/fdr-kg>

The repository includes trained model checkpoints, per-relation FDR statistics, and scripts for reproducing all figures and tables.

Data Availability

WN18RR and FB15k-237 are publicly available benchmarks. The specific train/validation/test splits used are the standard splits provided with the datasets. WN18RR is distributed by Dettmers et al. (2018); FB15k-237 is distributed by Toutanova et al. (2015).

Reproducibility

All experiments were conducted using Python 3.9 with PyKEEN 1.10.1, NumPy, and SciPy.

Model configuration: - Embedding dimension: $d = 256$ - Learning rate: 0.001 - Batch size: 256 - Training epochs: 20 - Random seed: 42 (all models)

FDR pipeline configuration: - Negative samples per test triple: $K = 100$ - Filtered setting: excluding all known true tails from training, validation, and test sets - Storey π_0 estimation: cubic spline extrapolation over $\lambda \in \{0.1, \dots, 0.9\}$ - Local FDR: Gaussian KDE with bandwidth selected via Silverman’s rule of thumb - Hardware: Apple M1, 16GB RAM

Runtime: - WN18RR (2,924 test triples): approximately 3–5 minutes per model - FB15k-237 (20,438 test triples): approximately 25–35 minutes per model

Extended Results

The following results are available in the code repository: 1. Full per-relation FDR statistics for all four models on WN18RR (TransE, RotatE, ComplEx, ConvE) 2. FB15k-237 per-relation FDR statistics (all 237 relations) 3. BH rejection counts at all FDR levels ($\alpha = 0.01, 0.05, 0.10, 0.20$) 4. Local FDR values for all individual test triples on both datasets 5. Sensitivity analysis results for additional K and λ values 6. q -value distributions for all models 7. Complete score files (positive and negative) for all 4 models on WN18RR

WN18RR Test Triple Exclusion

The WN18RR dataset contains 3,134 test triples. We retain 2,924 triples for FDR analysis. The excluded 210 triples (6.7%) involve head-relation pairs where fewer than 100 valid negative tail entities exist under the filtered setting, because nearly all candidate tail entities are known positives. This near-exhaustive positive coverage makes negative sampling infeasible and p -value construction unreliable. Following the standard filtered setting convention, these triples are excluded from the FDR pipeline; their exclusion does not affect comparative conclusions drawn across models, since all models are evaluated on the identical set of 2,924 triples.

FB15k-237 Per-Relation FDR Statistics

The complete per-relation FDR statistics for all 237 FB15k-237 relations under TransE, RotatE, ComplEx, and ConvE are available in the GitHub repository ([experiments/results/fb15k237/per_relation_fdr_full.csv](#)). Key aggregate findings (summarized in §4.6 of the main text):

- **Raw $p < 0.05$ rate range:** 0.8%–84.3% across 237 relations (TransE: 0.8%–72.1%; RotatE: 1.2%–84.3%)

- **Knowledge-blind relations:** 31 relations (13.1%) with near-zero raw $p < 0.05$ rates ($<5\%$) under both TransE and RotatE
- **Highly reliable relations:** 47 relations (19.8%) with $>50\%$ raw $p < 0.05$ rates under both models
- **Cross-model Spearman ρ :** >0.85 (relation difficulty is largely a property of the relation type rather than the model architecture)

A supplementary figure visualizing the per-relation reliability distribution across all 237 FB15k-237 relations is provided as `figures/fb15k237_per_relation_distribution.pdf`.

Multi-Seed FDR Stability Analysis

To assess the stability of FDR metrics across random initializations, we retrained TransE and ConvE under three random seeds each ($\{42, 123, 456\}$, 20 epochs, $K = 100$). Both models exhibit highly stable FDR metrics across seeds (coefficient of variation $< 2\%$ for $\hat{\pi}_0$, BH rejection rate, and MRR). Complete per-seed results and score files are available in the GitHub repository. See the main text (§4.7, Limitations) for a summary of these findings.

Per-Relation Sample Size Cautions

For WN18RR per-relation analysis (Table 2), three relations have small sample sizes: Relation 6 ($n = 26$), Relation 7 ($n = 22$), and Relation 8 ($n = 3$). FDR rejection rates for these relations should be interpreted cautiously. Relation 8 ($n = 3$, both models achieve 100% rejection) is included for completeness but is too small for reliable statistical inference.